



Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science & Information Technology

Mid Term Examinations (Fall 2022)

Course Code with Title	MS-104 : Differential Equation	Program	Bachelor of Computer Science BS(CS)
Instructors	Fariha Ansari/Maria Rashid	Semester	3 rd
Start date & Time	Dec 07, 2022 at 11:30 AM	End Time	01:00 PM
Maximum Marks	30		

IMPORTANT INSTRUCTIONS:

- Attempt All Questions.

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Question 1

(05+05+05+05)

(a) Let population of rabbits surviving at a time t be administered by differential equation $dP/dt = (P - 300)$ if $P(0) = 400$ then Solve the differential equation

(b) Solve the given Differential Equation $\frac{dy}{dx} + \frac{y}{x} = e^x$

(c) Solve the given initial value problem $xyy' = 2y^2 + 4x^2$; $y(2) = 5$

(d) Find the solution of the given Differential Equation $\cos x dx + \sin 2y dy = 0$

Question 2

(03+07)

Solve the following second order differential equations:

(a) $x^2 y'' - xy' + 2y = 0$.

(b) $y'' - 2y' + y = 0$; $y(0) = -1$, $y'(0) = 2$